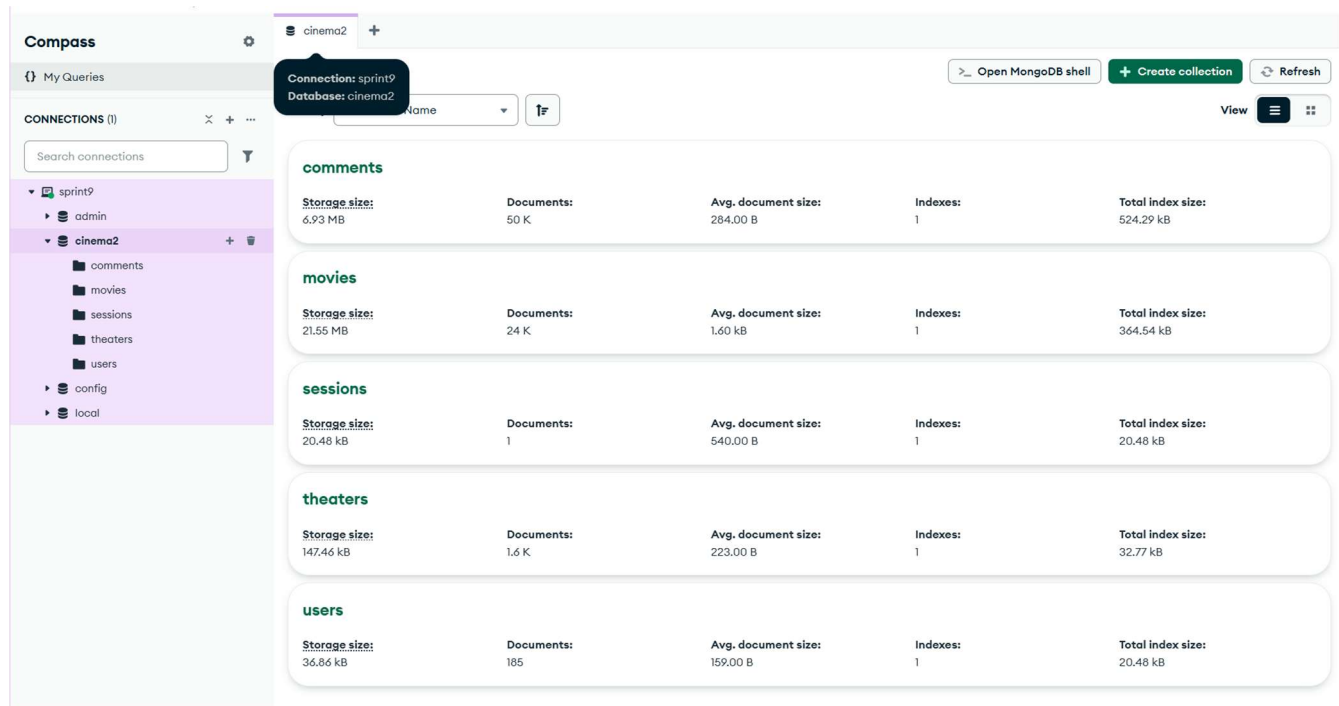


Nivell 1

Crea una base de dades amb MongoDB utilitzant com a col·leccions els arxius adjunts.



- Exercici 1

Mostra els 2 primers comentaris que hi ha en la base de dades.

```
>_MONGOSH
> use cinema2
< switched to db cinema2
> db["comments"].find().sort({date:1}).limit(2)
< {
  _id: ObjectId('5a9427648b0beebe695e67e'),
  name: 'Mercedes Tyler',
  email: 'mercedes_tyler@fakegmail.com',
  movie_id: ObjectId('573a1399f29313caabcec3ce'),
  text: 'Optio totam dolores magni. Enim ratione fuga tempora voluptatum est cumque animi. Quia nam doloribus corrupti
  date: 1970-01-01T01:07:09.000Z
}
{
  _id: ObjectId('5a9427658b0beebe6969efe'),
  name: 'Jon Snow',
  email: 'kit_harington@gameofthron.es',
  movie_id: ObjectId('573a13a4f29313caabd117df'),
  text: 'Dolorem animi tempora ullam quas. Iusto nobis reprehenderit aspernatur cupiditate a tempore. Commodi alias ex
  date: 1970-01-01T09:37:49.000Z
}
cinema2>
```

- Quants usuaris tenim registrats?

```
>_MONGOSH  
  
> use cinema2  
< switched to db cinema2  
> db["users"].countDocuments()  
< 185  
cinema2>
```

- Quants cinemes hi ha en l'estat de Califòrnia?

```
theaters  >_ mongosh: sprint9 +  
  
>_MONGOSH  
  
> use cinema2  
< switched to db cinema2  
> db.theaters.countDocuments({ "location.address.state": "CA" });  
< 169  
cinema2>
```

- Quin va ser el primer usuari/ària en registrar-se?

sprint9 > cinema2 > users

Documents 185 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or [Generate query](#) ✚

Project { field: 0 }

Sort { _id: 1 } Max Time MS 60000

Collation { locale: 'simple' } Skip 0 Limit 1

Index Hint { field: -1 }

ADD DATA EXPORT DATA UPDATE DELETE 25 1-1 of 1

```
{
  "_id": "ObjectId('59b99db4cfa9a34dcd7885b6')",
  "name": "Ned Stark",
  "email": "sean_bean@gameofthron.es",
  "password": "$2b$12$UREFwsRUoyF0CRqGNK0LzO0HM/jLhgUCNNIJ9RJAqMUQ74cr1J1Vu"
}
```

- Quantes pel·lícules de comèdia hi ha en la nostra base de dades?

```
>_MONGOSH

> use cinema2
< switched to db cinema2

> db.movies.countDocuments({ type: "movie", genres: "Comedy" });
< 7002

cinema2>
```

- Exercici 2

Mostra'm tots els documents de les pel·lícules produïdes en 1932, però que el gènere sigui drama o estiguin en francès.

The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'movies' and a '+' icon. Below it, the breadcrumb 'sprint9 > cinema2 > movies' is visible. The 'Documents' tab is selected, showing 23.5K documents. A query is entered in the search bar: `{ $and: [{ year: 1932 }, { $or: [{ genres: 'Drama' }, { languages: 'French' }] } ] }`. The interface includes buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Below the query bar, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. The results section shows two documents. The first document is for 'The Blood of a Poet' (1932), and the second is for 'Junta is hated by the people in the village where she lives, especialL...' (1932).

- Exercici 3

Mostra'm tots els documents de pel·lícules estatunidenques que tinguin entre 5 i 9 premis que van ser produïdes entre 2012 i 2014.

The screenshot shows the MongoDB Compass interface. The top navigation bar includes 'users', 'movies', and 'theaters'. Below it, the breadcrumb 'sprint9 > cinema2 > movies' is visible. The 'Documents' tab is selected, showing 23.5K documents. A query is entered in the search bar: `{ $and: [ { countries: 'USA' }, { 'awards.wins': { $gte: 5, $lte: 9 } }, { year: { $gte: 2012, $lte: 2014 } } ] }`. The interface includes buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. Below the query bar, there are buttons for 'ADD DATA', 'EXPORT DATA', 'UPDATE', and 'DELETE'. The results section shows one document for 'The Secret Life of Walter Mitty' (2013), which has 5 awards and was released between 2012 and 2014.

Nivell 2

- Exercici 1

Compta quants comentaris escriu un usuari/ària que utilitza "GAMEOFTHRON.ES" com a domini de correu electrònic.

```
>_MONGOSH
> use cinema2
< switched to db cinema2
> db.comments.countDocuments({ email: { $regex: "@gameofthron\\.es$", $options: "i" } });
< 22841
cinema2>
```

- Exercici 2

Quants cinemes hi ha en cada codi postal situats dins de l'estat Washington D. C. (DC)?

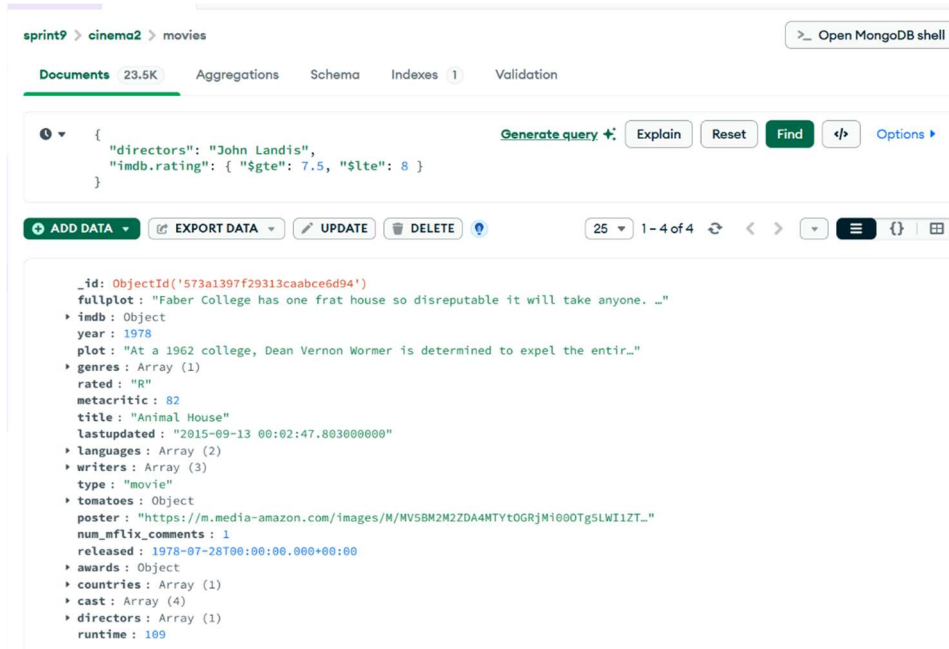
Código utilizado: `db.theaters.aggregate([ { $match: { "location.address.state": "DC" } }, { $group: { _id: "$location.address.zipcode", count: { $sum: 1 } } }, { $sort: { count: -1 } } ] )`

```
>_MONGOSH
> use cinema2
< switched to db cinema2
> db.theaters.aggregate([ { $match: { "location.address.state": "DC" } }, { $group: { _id: "$location.address.zipcode", count: { $sum: 1 } } }, { $sort: { count: -1 } } ] )
< [
  {
    _id: '20010',
    count: 1
  },
  {
    _id: '20002',
    count: 1
  },
  {
    _id: '20016',
    count: 1
  }
]
```

Nivell 3

- Exercici 1

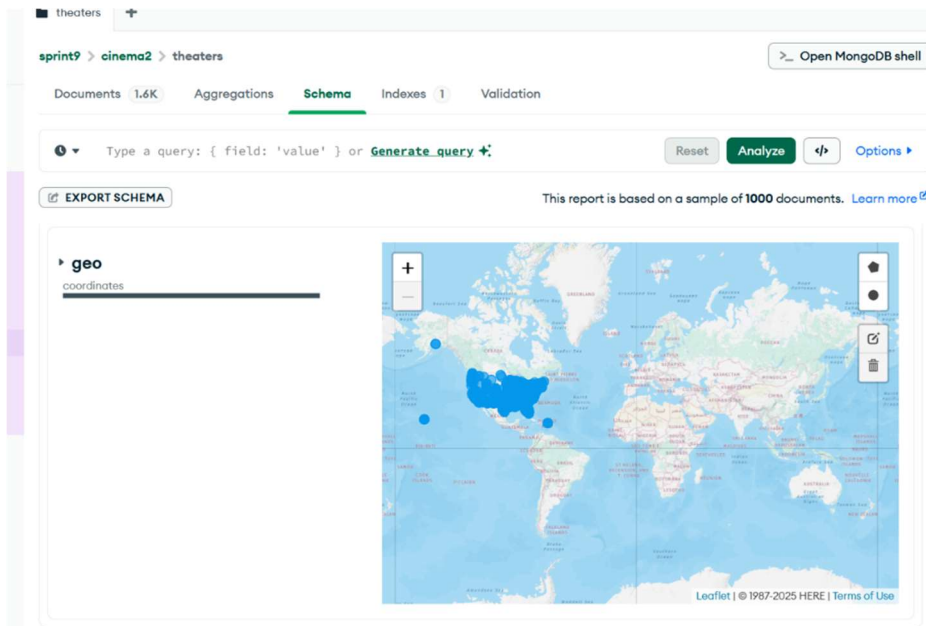
Troba totes les pel·lícules dirigides per John Landis amb una puntuació IMDb (Internet Movie Database) d'entre 7,5 i 8.



The screenshot shows the MongoDB Compass interface. The top navigation bar indicates the database is 'sprint9' and the collection is 'cinema2 > movies'. The 'Documents' tab is active, showing 23.5K documents. A query is entered in the search bar: `{ "directors": "John Landis", "imdb.rating": { "$gte": 7.5, "$lte": 8 } }`. Below the query bar, there are buttons for 'Generate query', 'Explain', 'Reset', 'Find', and 'Options'. The results section shows a single document for the movie 'Animal House' with the following details:
_id: ObjectId('573a1397f29313caabce6d94')
fullplot: "Faber College has one frat house so disreputable it will take anyone. ..."
imdb: Object
year: 1978
plot: "At a 1962 college, Dean Vernon Wormer is determined to expel the entire..."
genres: Array (1)
rated: "R"
metacritic: 82
title: "Animal House"
lastupdated: "2015-09-13 00:02:47.803000000"
languages: Array (2)
writers: Array (3)
type: "movie"
tomatoes: Object
poster: "https://m.media-amazon.com/images/M/MV5BM2M2ZDA4MTYtOGRjMi00OTg5LWI1ZT..."
num_mflix_comments: 1
released: 1978-07-28T00:00:00.000+00:00
awards: Object
countries: Array (1)
cast: Array (4)
directors: Array (1)
runtime: 109

- Exercici 2

Mostra en un mapa la ubicació de tots els teatres de la base de dades.



The screenshot shows the MongoDB Compass interface for the 'theaters' collection. The 'Schema' tab is active, showing the schema structure. A map visualization is displayed, showing the locations of theaters across the world. The map is titled 'geo' and shows a world map with several blue dots indicating theater locations. The map is interactive, with a search bar and a 'Generate query' button. The map is powered by Leaflet and OpenStreetMap data.